

MINERVA CryoCell — P&ID v2 Drawing Set Index

Project: MINERVA CryoCell — SCK CEN
Drawing No.: SCK CEN/84836013
Sheet size: A3 landscape (420 × 297 mm)
Standards basis: ANSI/ISA-5.1-2022, ISO 10628, IEC 60617, SCK CEN tag scheme `W-X:Y-Z-1`

1. Drawing split

The two original source drawings were each split into **two focused sheets** to improve readability. Each sheet is delivered in **two style versions**.

Source	Sheet	Focus
QCELL / LB	Sheet 1 — Cryogenic	40 K / 4.5 K / 2 K circuits + heat exchangers, vessels, valves
QCELL / LB	Sheet 2 — Instrumentation	Sensors, control loops, DIS interlock, signals
RFCELL (seen by ACR)	Sheet 1 — Process	DI-water / coupler process flow
RFCELL (seen by ACR)	Sheet 2 — Instrumentation	Sensors, control loops, DIS interlock, signals

2. Style versions

Version	Filename suffix	Description
B — Standard	<code>_STANDARD</code>	Balanced, full colour. Process lines primary (1.0 mm), equipment secondary (0.7 mm), bubbles 2 mm, signals 0.25 mm, tags 2.5 mm.
A — Control-Centric	<code>_CONTROL-CENTRIC</code>	Signals emphasised. Process de-emphasised to grayscale (0.5 mm), equipment background (0.35 mm), bubbles 3 mm, signals colour 0.5 mm, tags 3.5 mm.

Bubble diameters were scaled up from the 2 mm / 3 mm nominal to keep two-line ISA tags legible at the drawing's information density, while preserving the A > B emphasis relationship exactly.

3. File manifest (8 SVG + 8 PDF = 16 files)

output_v2/QCELL/

- QCELL-Sheet1-Cryogenic_STANDARD.svg / .pdf
- QCELL-Sheet1-Cryogenic_CONTROL-CENTRIC.svg / .pdf
- QCELL-Sheet2-Instrumentation_STANDARD.svg / .pdf
- QCELL-Sheet2-Instrumentation_CONTROL-CENTRIC.svg / .pdf

output_v2/RFCELL/

- RFCELL-Sheet1-Process_STANDARD.svg / .pdf
- RFCELL-Sheet1-Process_CONTROL-CENTRIC.svg / .pdf
- RFCELL-Sheet2-Instrumentation_STANDARD.svg / .pdf
- RFCELL-Sheet2-Instrumentation_CONTROL-CENTRIC.svg / .pdf

4. Companion documentation

- `01_LAYER_STRUCTURE_GUIDE.md` — the 14-layer toggleable hierarchy.
- `02_SENSOR_REALLOCATION.md` — sensor re-allocation mapping.
- `03_SCOPE_BOUNDARY_REFERENCE.md` — TPXYYYY scope-boundary reference table.
- `_build_meta.json` — machine-readable build metadata (re-allocations, scope codes).

5. Regeneration

```
cd generator && python3 build_pid_v2.py          # writes the 8 SVGs
python3 -c "import cairosvg,glob; [cairosvg.svg2pdf(url=f,write_to=f[:-4]+'.pdf') for
f in glob.glob('../output_v2/*/*.svg')]"
```

All geometry is reproduced from the structured segmentation data and the extracted symbol library; nothing is hand-placed bitmap.